mcidoc Student Documentation

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Table of Contents

1 Introduction

`mcidoc` is a LaTeX document class designed to provide students studying at **MCI Mechatonics** with all their documentation needs, whether it is to write a simple report or to write a their graduation thesis[1] whether it is a single or a double degree.

As it stands the following document types are supported and are implemented for both **english** and **german.**

- Thesis
- Report
- Paper (WIP)
- Notes (WIP)
- Slide (WIP)
- Poster (WIP)

All these aforementioned sub-classess will be explained in the following sections.

1.1 Purpose of Making

There are three (3) reasons for the creation of this class.

The first, is to standardise and streamline document creation which abides to the standards set by the MCI with regards to:

- Font-size, figure captions, bibliography, . . .
- Margin spacing, header and footer spacing, text information, . . .
- Spacing between lines, linebreak settings, spacing of headers, . . .
- Language determined grammatical rules (when to hypenate and caption text and proper quoatation marks)
- Simple error-correction codes to minimise wrong entering of values,

and so on so forth.

Its second purpose is to create a repository to house a **standardised** class to minimise any friction between the work of the student and the representation of the content.

> Care was given to how the class behaves to minimise its interaction with other packages the end-user might need for their work. The class only patches certain commands but never redefines them.

As a sidenote, `mcidoc` was also designed with customisation in mind by configuring its settings through the options presented in the `\documentclass` and using the macro `\SetMCI{}` which all its features are explained in this documentation.

The last purpose is perhaps more in line of streamlining. While LaTeX is an excellent typesetting language with is very powerful[2] capabilities, it is nevertheless a language with

[1] This is supported for both M.Sc and B.Sc degrees.

[2] LaTeX is a language known to be accidentally Turing complete (`https://beza1e1.tuxen.de/articles/accidentally_turing_complete.html`), meaning with enough time, it is possible to do all calculations any computer language can do. Therefore it is within the realm of possibility for one student to conduct their entire software work on LaTeX.

its quirks and customs that may not prove to be very useful outside of its own domain and therefore the user should not be worried of these quirks.

The user should only be responsible of the content they create. The form of the documentation, and how the content is presented is taken care of the class designer.

1.2 Reading the Commands and Environments

Throughout the document you will see code explanations as given below. Please read these options to get a better understanding of how the custom commands and environments are operated.

`\SetMCI` *marg key=val* [Command]

Allows global configuration of the document. Takes various `key=val` options based on the documentclass state and other sub configurations. Here the `key` is the configuation option, and `val` is the value it takes.

This command **must** be invoked **after** the `documentclass` but before `\begin{document}`.

For example, in the above the *Command* specifies that it is a LaTeX macro (i.e., `\SetMCI`) and that it takes a single **mandatory argument** (*marg*) with the value of `key=val`. Below this command description is a detailed information about the command and relevant information.

1.3 Currently Supported Formats

As mentioned previously the following `state`'s are supported

- **Thesis:** The primary way the thesis is written. Supports different options depending on the grade of the language requirements.

  To enable it, please write:

  ```
  \documentclass[state=Thesis]{...}
  ```

  where `...` described the path to the `mcidoc.cls`.

- **Report:** Used in creating reports for lectures and lab sessions. Examples include writing assignments, or lab reports.

  To enable it, please write:

  ```
  \documentclass[state=Report]{...}
  ```

  where `...` described the path to the `mcidoc.cls`.

All the aforementioned documents are bundled up as a class called `mcidoc`.

Important

This class has been tested on different computers running Windows and macOS with the complile being LuaTEX. However, as with everything, there will always be edge cases and I am sure to that end, if you encounter any error please create an issue in Github Repo (`https://github.com/dTmC0945/C-MCI-LaTeX-Class-mcidoc/tree/main`).

TEXnician's Note: This class is designed with LuaTEX in mind as it requires the use of the LATEX3 Kernel which comes precompiled with the engine. If one so wishes, they can force the use of `pdftex` to compile the document but additional packages would then be required for the compilation process.

Warning

While I will try to help with any problem this class might have, it is by no means I am providing any kind of WARRANTY or TECHNICAL SUPPORT. There will be documentation and some example documents which should compile without any additional configuration. Nevertheless, I can only help with the errors from code and not with your personal LATEX development environment (i.e., TEXStudio).

`state` [documentclass-value]

A configuration option to change the overall behaviour of `mcidoc` class. This value can take different options based on which formats are supported, which are shown at the beginning of this section.

The behaviour of the documentclass can be changed by setting `state=val` of the optional argument of the `mcidoc` documentclass.

2 Configuring the Document

The entire configuration is handled by a single command `\SetMCI` which the end user can change the behaviour by changing specific keys defined within. The accesible options changes depending on the `state` the `mcidoc` is in whereas some options are **global** and are accesible across all `states` of `mcidoc.cls`.

2.1 Notational Information

Before options for the `~\SetMCI{..}` are described, it is worth explaining the slight notational quirks used in defining the command.

For most, if not all, options given by `\SetMCI` and are by default either as `set` or `unset` unless specified otherwise within the documentation.

- If a feature is `set` by default, then the option will be executed as long as the user does not alter its value to `unset`.
- Conversely, if a `\SetMCI` value is `unset`, then it will not show in the document or alter anything within the document as long as the user does not change its behavior.

PROPERTY	**RESULT**
`set`	Enable a feature
`unset`	Disable a feature

Some options on the other hand can take different values which are explained in detail where they are relevant.

Throughout the documentation, various configuration to the `\SetMCI` command is given in the form of `/root/parent/child` for clarity.

For example the `/Thesis/Supervisor/Postfix` option can be invoked in the documentation as such:

```
...
\SetMCI{
  Thesis = {
    Supervisor = {
      Postfix = {},
    },
  },
}
...
```

where `...` designates the code prior and after the `\SetMCI` command, and `{}` is where the value is entered.

Alternativey, the option `/Degree` can be set as follows:

```
...  \SetMCI{ Degree = {}, } ...
```

where again, `...` designates the code prior and after the `\SetMCI` command, and `{}` is where the value is entered.

2.2 Global Options

The following options are available across different states:

TEXnician's Note: These options are defined as `\Global@` prefix.

`/Degree` *marg* [key]

Sets the **Degree** of the document which can change some subtitles of the documentation.

Current supported values are: `BSc`, `MSc`.

`/Language` *marg* [key]

Sets the **Language** of the document which changes the typographical rules of the document and correctly sets the document titles and labels.

Current supported values are: `EN` (english), `DE` (german).

`/Deparment` *marg* [key]

Sets the **Deparment** of the document.

Current supported values are: `MECH` (Mechatronics).

`/StudentID` *marg* [key]

Sets the **Student Number** (i.e., Matrikelnummer) of the document.

As it is a FH, the number **must** begin with the digit `5`.

2.3 Thesis Specific Options

The following options are used in the thesis mode only. Some options may have the same name as other states. In those cases, these options will only be described under the parent option in which they were invoked.

The following are the global options for the `Thesis` state:

`/Thesis/Title` *marg* [key]

Set the title for the Thesis.

`/Thesis/StudyProgram` *marg* [key]

Set the study program for the Thesis.

`/Thesis/Location` *oarg* [key]

Set the location in which `Preclusion from Public Access`, and `Declaration In Lieu of Oath` are signed.

Default is `Innsbruck`.

`/Thesis/Date` *oarg* [key]

Set the date in which `Preclusion from Public Access`, and `Declaration In Lieu of Oath` are signed.

Default is the date in which the document is compiled.

To change it the user can enter a date in the form `{DD.MM.YYYY}`.

`/Thesis/Embargo` *optional* [Option]

Set the end date for the embargo of the thesis.

The entered date must be in the format `dd.mm.yyyy`.

Default is `unset`.

2.3.1 Custom Environments and Commands

There are a few commands and environments defined specifically for the `Thesis` class which are as follows:

`Abstract` *marg:language* [Environment]

Generate an abstract after **Declaration in Lieu of Oath** if **Acknowledgements** is not inserted, else the abstract is inserted right after **Acknowledgements**.

The abstract(s) is/are printed before the main Table of contents.

The mandatory argument `{language}` controls which language is used. Currently `EN` and `DE` are supported. Based on the chosen `/Language`, The following actions are taken.

1. If English is chosen (`EN`), then only the English abstract is printed with the title **Abstract**.
2. If German is chosen (`DE`), then first the German abstract with the title **Zusammenfassung** is printed, then the English abstract is printed with the title **Abstract** is printed.

NOTE: This is a **mandatory** environment.

TeXnician's Note: Abstracts are written to external files with the name `aux-\jobname-abstract-<lang>.abs` where `<lang>` is the lowercase of `/Language`. The printing of this is achieved by the `\VerbatimOut` to preserve the user entered macros.

`Acknowledgments` [Environment]

Generate the **Acknowledgements** page for the thesis. This is printed right before any abstract on the thesis.

NOTE: This is an **optional** environment.

TeXnician's Note: The **Acknowledgments** are written to external files with the name `aux-\jobname-acknowledgement.ack`. The printing of this is achieved by the `\VerbatimOut` to preserve the user entered macros. To control whether the acknowledgement is to be printed a globally defined `\ifacknowledgments` is used. This must be set global to allow it to be seen by the internal command `\Th@Preamble`.

2.3.2 Supervisor

Supervisor is the primary person when the student is conducting their thesis. This person can be an internal (i.e., MCI Lecturer, Researcher) or could be someone from a company.

An example of this options is shown as below:

```
...
Supervisor     = {%
  Company = {Ministry of Sillyputty}, % <- The company of the supervisor
  Prefix  = {Dipl-Eng.},
  Name    = {Colin Devon},
  Postfix = {M.Sc},
},%
...
```

`/Thesis/Supervisor/Company` [key]

Inserts the company name which the supervisor belongs to. This option should only be used if the supervisor is **outside** MCI as using this option will require the insertion of the Section 2.3.3 [Reviewer], page 7, option as well.

Default is `unset`.

`/Thesis/Supervisor/Prefix` [key]

Prints prefix honorifics for the supervisor, if there is any. For example titles such as Dr. or Dipl-Eng should be added here.

Default is `unset`.

`/Thesis/Supervisor/Name` [key]

Prints the name of the supervisor.

Default is `unset`.

`/Thesis/Supervisor/Postfix` [key]

Prints post honorifics of the supervisor. For example, titles such as Ph.D should be added here

Default is `unset`.

2.3.3 Reviewer

Reviewer is the examiner when the student is conducting their thesis with a company. The reviewer can be considered as a liason between the student and the MCI.

An example usage of this is shown below

```
...             % <- Previously the Supervisor was
...             % defined so we add the Reviewer
Reviewer     = {%
     Prefix     = {Rev.},
     Name       = {Ted Marshall},
     Postfix    = {Esq.},
   },%
...
```

`/Thesis/Reviewer/Prefix` [key]

Prints prefix honorifics for the Reviewer, if there is any. For example titles such as Dr. or Dipl-Eng should be added here.

Default is `unset`.

`/Thesis/Reviewer/Name` [key]

Prints the name of the reviewer.

Default is `unset`.

`/Thesis/Reviewer/Name` [key]

Prints post honorifics of the reviewer. For example, titles such as Ph.D should be added here

Default is `unset`.

2.3.4 Table of Contents

Allows the control of how the table of contents is printed.

`/Thesis/ToC/Primary` [Variable]

Prints the main Table of Contents. Default is `set` and is the first table of content to be printed right after the abstract.

Default is `set`.

`/Thesis/ToC/Figures` [Variable]

Prints the **List of Figures** right after Table of Contents on a newpage.

Default is `set`.

If user has no figures within their document, this should changed to `unset`.

`/Thesis/ToC/Tables` [Variable]

Prints the **List of Tables** right after **Contents** on a newpage if `/Thesis/ToC/Figures` is `unset` or right after `/Thesis/ToC/Figures` and on the same page if space if available.

Default is `set`.

If user has no table within their document, this should changed to `unset`.

2.3.5 Example: Double Degree Programme

The following setting is defined for the student `Karl Hendrix` doing a double degree with ASU (Appalachian State University). An example `\SetMCI` option would be as follows:

```
\SetMCI{%
  StudentName    = {%
    Name = {Karl Hendrix}
  },%
  StudentID      = {52414238},
  Degree         = MSc,
  Language       = EN,
  Department     = MECH,
  Thesis = { %
    Title          = {The Effects of Amplifier Gain on Guitar Pedal Distortion},
    StudyProgram   = {Smart Technologies},
    Supervisor     = {%                    % <- The MCI Supervisor
      Prefix  = {Prof.},
      Name    = {David R. M. Jazz},
      Postfix = {},
    },%
    Double-Degree  = {
      Institution     = {ASU},             % <- The University code for ASU
      Student-Number  = {6712345},
      Supervisor     = {%                  % <- Supervisor at ASU
        Prefix  = {Prof.},
        Name    = {Colin Stevens},
        Postfix = {},
    },%
```

```
      },
      Embargo        = {31.07.2031},         % <- An embargo request
      Bibliography   = {%
        Location     = {references.bib},     % <- Bibliograpy resides in the root
      },
    },
  }
```

2.3.6 Example: Company Based Thesis

2.3.7 Example: BSc Thesis in English

2.3.8 Example: MasterArbeit auf Deutsch

3 Preparing a Thesis

3.1 Writing the Abstract(s)

Depending on the language requirements the thesis must either have:

- An abstract in English (`EN` only)
- An abstract in German and then in English (`DE` only)

For example the following environment generatex an abstract for the english section of the astract.

```
\begin{Abstract}{EN}

  In recent years with the improvements in power electronics the use
  of high speed applications have increased. The industry standard
  Caged-Rotor Induction Motor (CRIM) is not suitable for high speed due
  of its mechanical durability and thermal properties. The most common
  rotor type for this application is a Solid Rotor Induction Motor
  (SRIM). This class of rotors are known to have high mechanical
  durability and good thermal properties.

  This project focuses on comparing a CRIM with 3 types of SRIM. These
  rotors will be tested on a induction motor (IM) standard stator. The
  windings will be changed in order to keep the current passing in the
  stator for each rotor constant. The CRIM and smooth type SRIM will be
  testedn in İTÜ Elektrik Makinaları Laboratory and their high speed
  applications will be compared.

\end{Abstract}
```

4 Preparing a Lab Report

5 Preparing a Presentation

6 Command Index

Below are the index entries of all relevant `mcidoc` commands, keys, and arguments.

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